

QC analysis of raw sequencing data

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Introduction

In this RMarkdown the FastQC and multiQC analysis of the .fastq-files from [GSE138852](#) is described. These .fastq-files contain the raw sequencing data and the phred-scores generated on the Illumina NextSeq 500 high-output lane with the 10x Genomics Chromium Single Cell 3' Library & Gel Bead Kit v2. This markdown explains how analysis should be done and what the conclusion and discussion is in regards to the research question.

Research question

For this project the main research question is *“How can a reliable and reproducible analysis pipeline be established to identify cell-type-specific differential gene expression from snRNA-seq data in Alzheimer’s disease?”*, the first step in answering this question is looking at the quality of sequencing data. The sub-question that is answered in this RMarkdown is as followed: Do the sequencing reads have a high enough Q-score to further analyse the data ($Q \geq 30$)?

Packages and Requirements

In order to run the scripts for QC-analysis a few packages need to be installed first, for analysis an environment was created in miniconda with the command line prompt;

```
conda create -n snrnaseq
```

The conda environment was activated with the prompt

```
conda activate snrnaseq
```

Install the required packages in the conda environment. For obtaining the fastq-files and analyses the following packages are required:

- sra-tools
- fastqc
- multiqc

Run the following commands in the terminal to install the packages:

```
conda install bioconda::sra-tools
conda install bioconda::fastqc
conda install bioconda::multiqc
```

Obtaining the data

The first step in the data-analysis pipeline is obtaining the data. The accession numbers of the GSE138852 dataset that need to be downloaded are listed in the text file *SRR_Acc_List.txt*. To download all the fastq files a bash script is written (*01_download_SRR_fastq.sh*), this script uses the *sra-tools*-package. The first section of this script defines the location where the *SRR_Acc_List.txt* is stored and an output directory, if this is non existing the script will create the output directory prior to fetching the files. After creating an output directory the script uses the *SRR_Acc_List.txt* and reads the first accession number that starts with SRR, it downloads the accession number with *fastq-dump* from the GEO database and stores it in a compressed format (gzip) to the output directory. After downloading a file it searches for the next accession number in the textfile and repeats the cycle.

The script takes a long time since the downloads are quite large, therefore it is recommended to add *nohup* and *&* to the command line to run the script on the background of the terminal. Navigate to the scripts folder and run this script from the terminal with the following command:

```
nohup bash 01_download_SRR_fastq.sh &
```

Analysis with FastQC

For FastQC analysis the downloaded fastq-files are used as input data. The fastq files contain the sequencing reads and the quality score (Q-score or Phred score) that are associated with those reads. For each read there are four lines:

1. The first line is the identifier, this is the ID that is generated by the sequencer.
2. The second line contains the nucleotide sequence.
3. The third line is the symbol "+", which separates the sequence from the Q-score
4. The fourth line contains the Q-score this is given as a character from the ASCII table.

Analysis of the fastq files is done using a script (02_fastqc.sh) that analyzes all fastq files one after the other. This script uses the fastqc-package to create a report, the output of this script are two files:

- A html report, which contains a variety of plots such as the per base sequence quality and per sequence quality scores.
- A zipfile that contains the raw data for each plot in text format.

To run this script it is recommended to add nohup and & to the command line since there are quite a lot of files to analyse. Navigate to the scripts folder and run this script from the terminal with the following command:

```
nohup bash 02_fastqc.sh &
```

The html report will automatically be saved to the output folder that was specified in the script, two examples of the plots that were analysed were visualised in figure 1 and figure 2. In figure 1 an example is shown of the plot that is generated for the per base quality score. On the x-axis the position of the base in the read and on the y-axis the quality score is shown. The colours on the background of the plot are an indication of the quality, with green being a very reliable quality, and red being of less reliable quality.

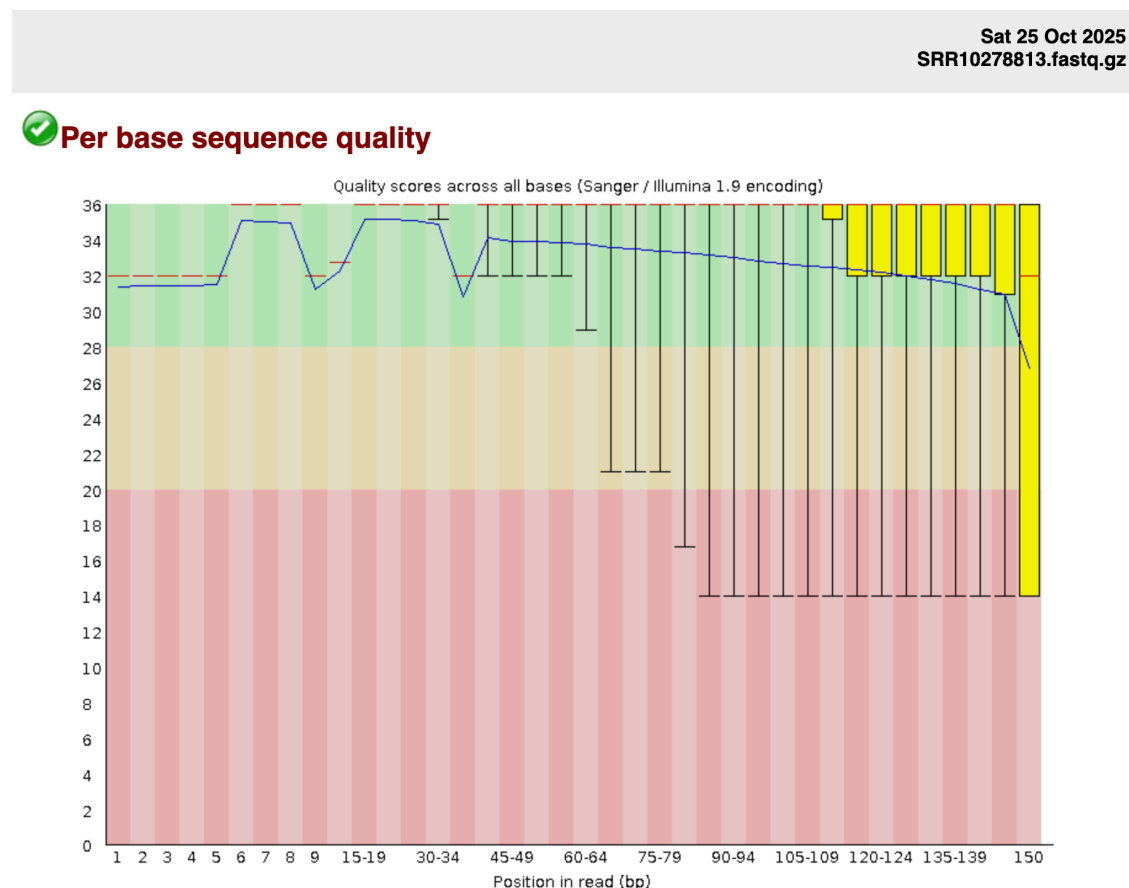
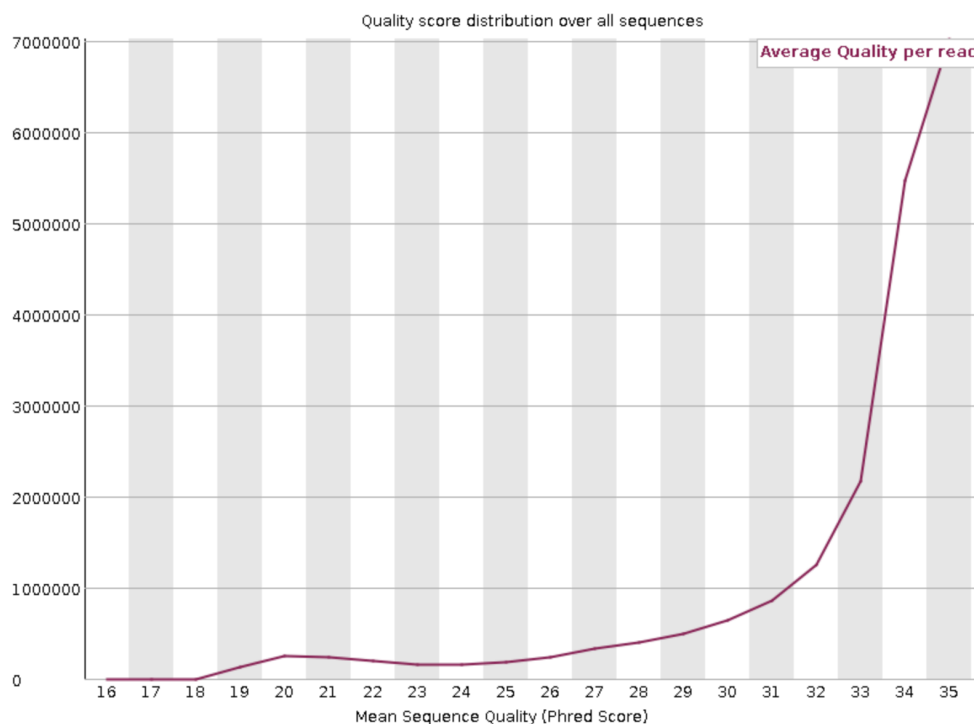


Figure 1: Per base quality score. This figure shows the per base quality score of the data of accession number SRR10278813. This plot was generated by fast QC.

In figure 2 an example is shown of the plot that is generated for the per sequence quality score. On the x-axis the mean quality score is shown and on the y-axis the number of sequences with that quality score.

✔ Per sequence quality scores



Analysis with MultiQC

The analyses of all the plots that were generated in **analysis with FastQC** is quite time consuming. To optimize this part of the workflow the package **MultiQC** was used. No script was written for MultiQC, after installing multiQC it is possible to run MultiQC from the command line.